

NEXORA

FogNet SafeEnvelope SafeCorridor Edition

A radar-first system that proves the next stopping distance is clear before a mine vehicle commits to it

SIH26-26007

Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines

Master solution dossier - source audit, competitor comparison, selected architecture, original research modules, algorithms, implementation, cost, validation, safety case, SIH demonstration and deployment gates.

Prepared from Nexora's submitted FogNet PDF/PPT/diagrams, the user-supplied Trinetra reference concept, ISO and mine-industry guidance, adverse-weather sensing research, product documentation and Indian compliance sources.

Version 3.0 | 19 September 2026

DECISION SUMMARY

Read this first

Nexora already has the correct foundation: radar-first obstacle ranging, a dynamic safe-speed model, roadside visibility sensing, cooperative awareness, an offline local loop and degraded operation. The winning move is not to add more sensors. It is to turn those parts into one auditable decision that competing dashboards do not show.

Selected idea

Safe-to-Proceed Corridor: the vehicle continuously proves a clear route length ahead and compares it with a conservative, vehicle-specific stopping requirement. If the proof is shorter than the stopping requirement, the system recommends a lower speed **before** the vehicle enters the untrusted zone.

Layer	Nexora decision
Primary safety loop	77 GHz radar -> tracked hazards -> clear corridor -> stopping envelope -> local HMI
Fog	Measure current MOR; provide a calibrated trend/probability only after site data exists
Blind curve	Fail-closed Virtual Block token plus local radar; cooperative data never clears a local hazard
GNSS weakness	Road graph + IMU and optional passive RadarGlyph landmarks at critical zones
Braking	Conservative BrakeTwin by vehicle/load/grade/surface; evidence may reduce speed immediately but never raise it without review
Communication	Fresh, signed, route-relevant peer messages; LoRa only for low-rate fog telemetry
Production control	Advisory first; OEM and ISO path before retarder or brake intervention

Truth boundary

No responsible report can guarantee that an idea has never been considered anywhere. This dossier identifies a **distinctive combination** and testable research hypotheses. Patent novelty and freedom-to-operate require a professional prior-art search. Prototype success is not production safety certification.

NAVIGATION

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PROBLEM FRAMING

1. The real problem and the engineering success condition

The SIH statement concerns safe and efficient movement of mine vehicles when dense fog reduces visual range on open-cast iron-ore haul roads. The challenge is not merely image enhancement or obstacle detection. A loaded dumper must avoid vehicle-to-vehicle, vehicle-to-person and vehicle-to-environment conflicts while its driver, optical sensors, road friction, localization and communication may all degrade together. [R1, R2]

The unsafe condition

A collision becomes possible when the distance the vehicle can trust as clear is shorter than the distance needed to perceive, decide and stop. This gives Nexora a precise engineering success condition:

Verified clear corridor $\geq$ reaction distance + braking distance + uncertainty reserve.

Failure source	Why ordinary solutions fail	Required response
Dense fog	Human vision and cameras lose contrast; a fog percentage is not a stopping decision	Measure MOR, gate optics by health, keep radar as primary range input
Wet/downhill road	Fixed 10 m or 20 m alarms ignore actual stopping ability	Use load, grade, wetness, delay and tested deceleration
Blind curve	Neither vehicle radar can see through terrain	Manage route occupancy before entry; use peer path only when fresh
GNSS drift	Map icon can appear on the wrong branch or bench	Expose pose uncertainty; cap speed; use local anchors at critical zones
Network loss	Cloud/V2V-only systems lose awareness	Local safety loop remains independent; remote services degrade gracefully
Alarm fatigue	A crowded screen and repeated beeps are ignored	One priority, one safe speed, one direction and one reason

Operational Design Domain to freeze before coding

- Named vehicle class, loaded/unloaded mass range, tires, brake state and approved maximum speeds.
- Surveyed route graph, travel direction, grades, widths, blind segments, intersections, berms and stop lines.
- Tested limits for fog/MOR, rain, dust, darkness, temperature, vibration, sensor contamination and GNSS/network availability.
- Advisory functions only for SIH and first mine pilot; no automatic service-brake claim.

WHAT TO KEEP, CORRECT AND REMOVE

2. Detailed audit of Nexora's current FogNet solution

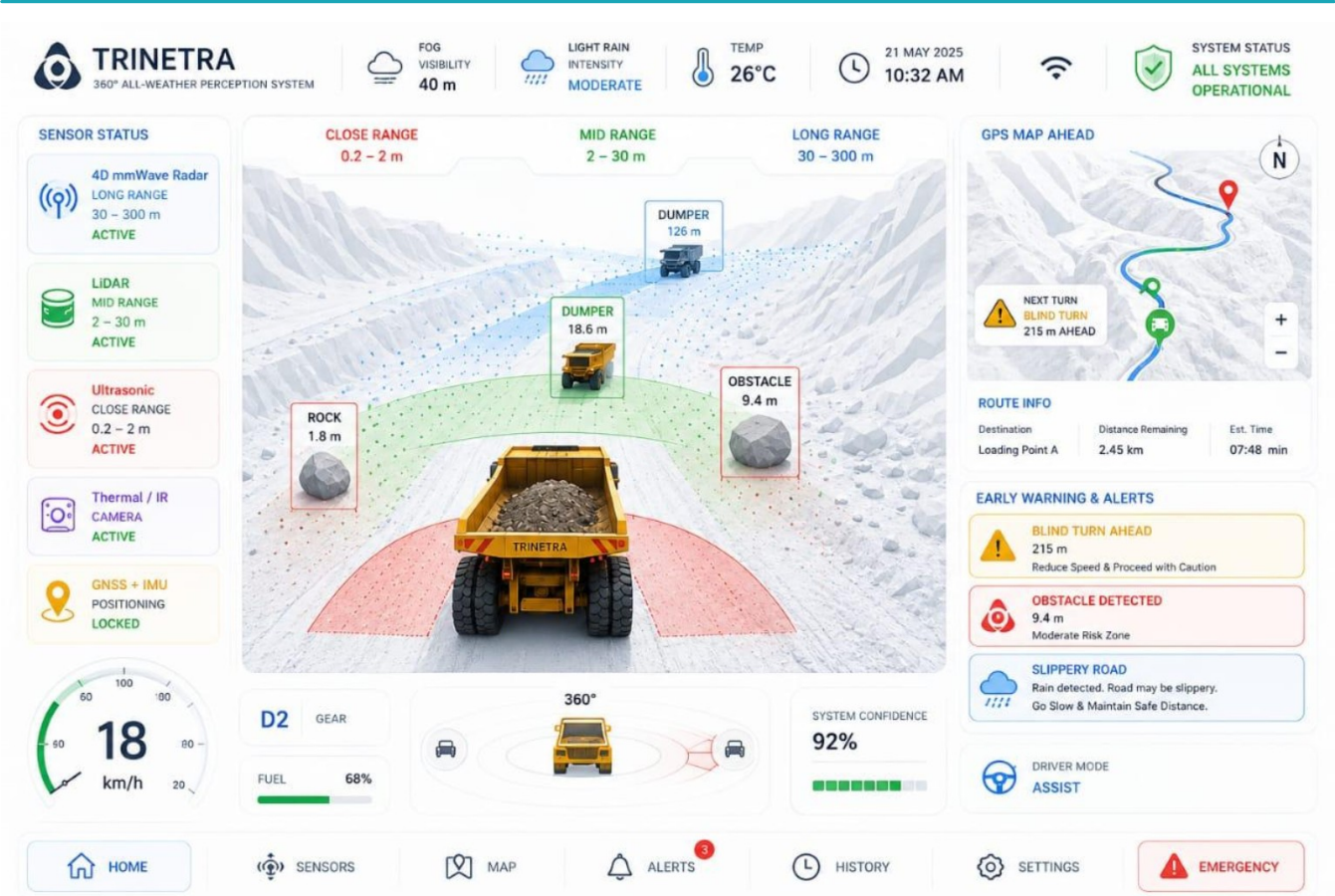
The source PDF describes fixed fog stations, radar/thermal/NIR/RGB perception, GPS/IMU, LoRa V2V, confidence-weighted fusion, safe-speed calculation, automatic stopping, an Alpha escort mode, a dashboard and event logging. The PPT adds radar-first detection, MOR monitoring and fail-safe operation. The diagrams show a workable sense-decide-alert-share flow. The following audit preserves the valuable system thinking while removing claims judges can challenge.

KEEP	Reason
Radar as primary ranging	Appropriate for adverse visibility, provided range and contamination are validated in the declared environment. [R8, R9]
Dynamic safe speed	Much stronger than a fixed proximity zone when tied to measured braking, route and uncertainty.
MOR roadside sensing	Measures optical visibility directly; humidity is supporting context, not the primary truth. [R7]
Offline-first local loop	A mine safety alert cannot depend on cloud or public mobile coverage.
Tiered multimodal HMI	Useful when the number of states, reasons and recovery rules are controlled.
Event logging	Required to reproduce decisions, tune nuisance alerts and build a safety case.

CORRECT OR REMOVE	Why	Replacement
Guaranteed 10-20 min fog prediction	No site dataset or calibrated model supports it	Report current MOR and trend; add probabilistic nowcast only after validation
LoRa as continuous V2V safety link	LPWAN delivery and airtime are not a braking guarantee	Use direct/site link for peer awareness; keep LoRa for sparse MOR telemetry
Automatic stop in student prototype	Unsafe without OEM interface, functional-safety analysis and vehicle tests	Advisory plus mock relay/scale cart; production path under ISO 21815 and ISO 19014 [R3-R6]
AI fusion as final decision	Opaque score cannot own a safety boundary	Use AI for classification/nowcast; deterministic stopping envelope owns alert state
AMG8833 as long-range person detector	8 x 8 thermal array cannot support that claim	Near-field demo only, or optional higher-resolution module with range test
Alpha escort as failover	Following another vehicle while one's own primary sensor is failed can increase exposure	Stop in a safe place or restricted recovery procedure defined by mine management
Supervisor instant stop broadcast	Remote command may be delayed/spoofed and bypass OEM controls	Authenticated advisory/emergency message; no direct production brake actuation

USER-SUPPLIED COMPARISON IMAGE

3. Review of the other team's Trinetra reference concept



Trinetra reference image supplied by the user. Its displayed specifications and performance claims have not been independently verified.

What the image communicates well

A polished 360-degree perception dashboard, active sensor health, close/mid/long range zones, obstacle labels, route context, weather, driver alerts and a system-confidence indicator. It is easy for a judge to understand at first glance.

COMPETITIVE STRATEGY

4. Trinetra comparison: where Nexora must be different

Trinetra's visible feature set overlaps Nexora's earlier architecture: radar, LiDAR, ultrasonic, thermal, GNSS/IMU, mapping and alerts. Therefore, winning by listing the same sensors is unlikely. Nexora should win on the decision guarantee, failure behavior and evidence.

Question a judge may ask	Reference concept shows	Nexora should prove
What distance is actually safe?	Fixed sensor zones and an obstacle at 9.4 m	Vehicle-specific stopping envelope and remaining reserve
Why is confidence 92%?	One aggregate confidence number	Per-input health, age, uncertainty and a visible reason for the final state
What happens around terrain?	Map and blind-turn warning	Fail-closed segment token before entry plus local radar inside
What if GNSS is wrong?	Position locked	Pose covariance, map-consistency test and optional radar landmark anchor
What if network disappears?	Operational Wi-Fi icon	A demonstrated independent local loop and queued evidence
How do sensors remain trustworthy?	Active/inactive status	Contamination detection, depot reflector self-test and watchdog
Is it deployable?	Comprehensive HMI	Advisory pilot, acceptance metrics, FMEA, maintenance and OEM gates

Illustrative stopping check from the screenshot

At 18 km/h, speed is 5 m/s. With an illustrative 1.0 s response delay and 1.5 m/s² effective deceleration, reaction distance is 5.0 m and braking distance is about 8.3 m, totaling 13.3 m before any buffer. An obstacle displayed at 9.4 m could therefore already be inside the required envelope. The lesson is not that Trinetra is unsafe - its internal model is unknown - but that **range zones and object boxes alone do not prove a safe speed.**

Nexora's one-line distinction

Other systems show what the sensors detect. **Nexora shows whether the complete stopping corridor is proven clear - and why.**

SELECTED CONCEPT

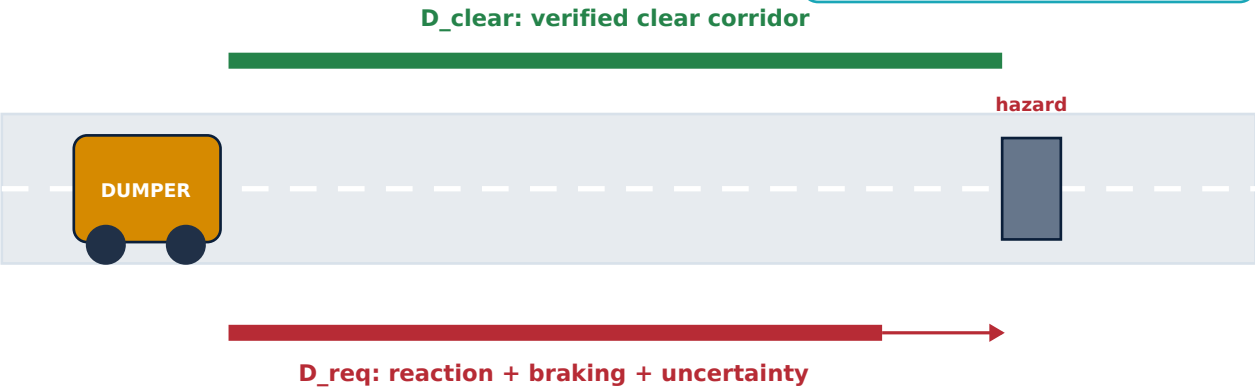
5. The central innovation: Safe-to-Proceed Corridor

THE CORE DECISION

Proceed only while $D_{clear} \geq D_{req} + \text{reserve}$.

$$\text{Margin} = D_{clear} - D_{req}$$

positive = safe reserve



Unlike a fixed red/yellow/green range zone, the boundary moves with payload, grade, wetness, delay and evidence quality.

Four corridor boundaries

Boundary	Evidence	Conservative behavior
Nearest route conflict	Radar track, mapped hazard or validated peer path	Use earliest conflict interval, not nearest Euclidean dot
Validated sensor horizon	Tested radar range at current health and mounting	Reduce when radome is dirty, clutter rises or diagnostics fail
Route/map horizon	Surveyed road graph and current branch confidence	Stop corridor at blind geometry, closure or ambiguous branch
Localization horizon	Pose covariance propagated along route	Increase buffer and cap speed as uncertainty grows

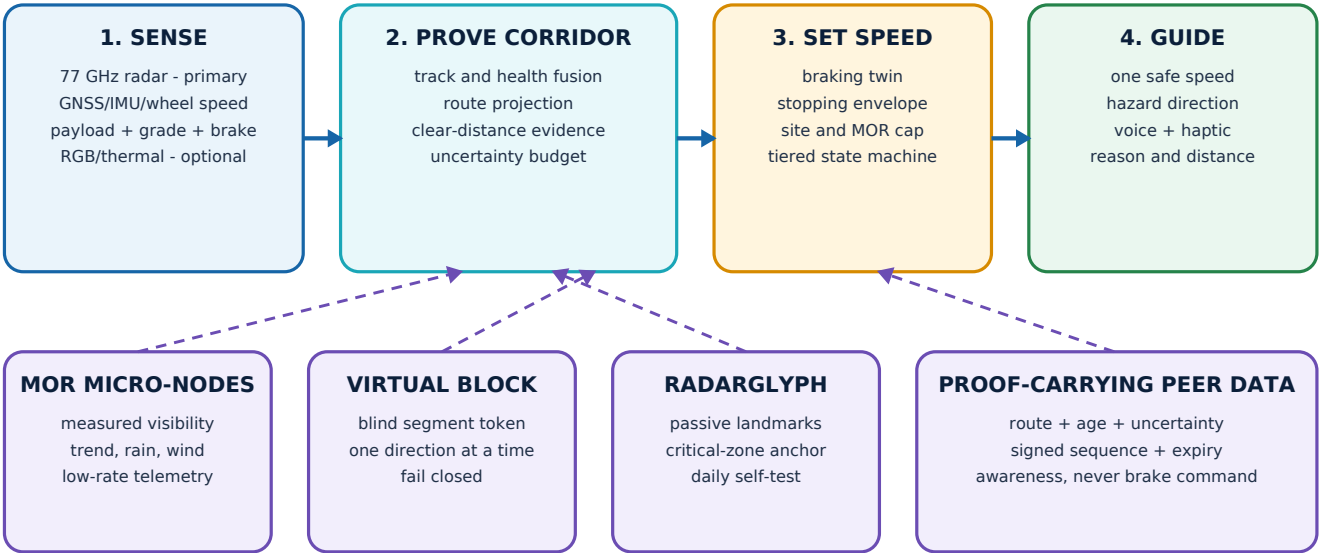
Unknown-space debt

Define **unknown-space debt** = $\max(0, D_{req} - D_{clear})$. A positive value means the vehicle needs more verified road than it currently has. The HMI should translate that into a lower recommended speed and reason such as 'blind curve proof missing' or 'radar range reduced'. This is more meaningful than a generic 92% confidence score.

ONE SAFETY LOOP, SUPPORTING INTELLIGENCE

6. Recommended end-to-end architecture

NEXORA SAFE-TO-PROCEED ARCHITECTURE



Safety invariant: loss of network, camera, dashboard or fog node cannot suppress a valid local radar alert.

Safety ownership

Owner	Can raise risk?	Can lower risk?	Can command production brake?
Local radar and track health	Yes	Only with fresh validated evidence	No in SIH/advisory build
SafeCorridor engine	Yes	Yes within frozen rules and ODD	No in SIH/advisory build
MOR/weather node	Yes, by reducing visibility/site cap	No automatic increase without stable recovery	No
Peer/V2V message	Yes, adds a conflict	Never clears a local radar hazard	No
Control room	May issue advisory/closure	Cannot remotely suppress a local critical alert	Only through later approved OEM system
Camera/thermal	May confirm class or add evidence	Cannot veto radar solely because it sees nothing	No

Non-negotiable invariant
A valid local radar conflict must survive loss of camera, fog node, V2V, map server, dashboard and internet. Supporting systems may make the vehicle more cautious; they must not silently make it less cautious.

PURPOSE-DRIVEN, NOT SENSOR-HEAVY

7. Hardware responsibilities and sensor selection

Component	Owns	Does not own	Prototype note
77 GHz radar	Primary range, relative speed, azimuth and object tracks	Reliable semantic class in every condition	Use 60 GHz evaluation board for scale proof only; production range must be tested
GNSS + IMU + wheel speed	Pose, motion and route match with covariance	Obstacle detection	Use logged replay before RTK; add RTK only if route ambiguity requires it
Payload/grade/brake inputs	Stopping model context	Collision detection	Manual payload mode and surveyed slope acceptable for SIH
MOR node	Current optical visibility and trend	Object detection or guaranteed 20-minute prediction	Calibrate against a known path/reference
RGB/NIR camera	Evidence, recording and classification in usable visibility	Safety range in dense fog	Borrowed camera is sufficient
Thermal	Optional worker/night evidence	Precise long-range distance by itself	Omit unless worker scenario is central
Safety MCU	Watchdog, HMI heartbeat and safe output state	Complex perception	Separate from Linux compute
Edge computer	Fusion, route projection, logging and dashboard feed	Certified brake control	Raspberry Pi 5/mini PC for demo

Why not LiDAR in the core?

LiDAR can be useful for clear-weather mapping, but fog and suspended particles can degrade optical ranging. Radar is the primary safety-ranging choice because the target ODD is low visibility. Radar is not magic: heavy rain, multipath, contamination and low-reflectivity targets still require characterization and health monitoring. [R8, R9]

Daily self-test

- At power-on: firmware/config signature, map version, clock, radar link, GNSS/IMU, HMI, MCU watchdog and storage.
- At depot: approach a surveyed reflector/target; check range, angle and signal-strength residual against the allowed band.
- During shift: monitor blocked radome/lens, stuck readings, impossible jumps, CPU deadlines, packet age and time sync.

DETERMINISTIC DECISION

8. Safe-speed mathematics that judges can audit

For an available verified distance D_{avail} , use a conservative response delay t_{delay} and effective deceleration a_{eff} . Define downhill grade as positive. A simple longitudinal model is:

Stopping requirement

$$a_{\text{eff}} = a_{\text{brake, lower}} - g \sin(\text{grade}) - a_{\text{surface_penalty}}$$
$$D_{\text{req}} = v t_{\text{delay}} + \frac{v^2}{2 a_{\text{eff}}} + D_{\text{buffer}} + D_{\text{uncertainty}}$$

Maximum advised speed

$$v_{\text{safe}} = a_{\text{eff}} [-t_{\text{delay}} + \sqrt{t_{\text{delay}}^2 + 2 D_{\text{avail}} / a_{\text{eff}}}]$$

Then apply: **$v_{\text{cmd}} = \min(v_{\text{safe}}, \text{site limit}, \text{visibility cap}, \text{curve cap}, \text{confidence cap})$.**

Inputs and fallbacks

Input	Preferred source	If missing or stale
Payload	Vehicle CAN/load system	Use worst validated loaded class
Grade	Surveyed route plus IMU	Use worst local downhill grade plus margin
Deceleration	Approved BrakeTwin bin	Use minimum approved class value
Surface	Rain/road input plus historical evidence	Assume wet/low-friction bin
Response delay	Measured sensor-to-driver chain and human-factors allowance	Use frozen conservative upper bound
D_clear	Route-projected radar/peer/map evidence	Shorten at first untrusted boundary

Do not use raw TTC alone

Time-to-collision becomes uninformative near zero relative speed and does not cover a stationary obstacle, blind horizon or available stopping distance. TTC may be displayed as supporting context; the stopping margin owns the state.

DIFFERENTIATOR 1

9. BrakeTwin: a conservative vehicle-specific braking model

Two dumpers at the same speed can need different stopping distances because of payload, downhill grade, tires, brake temperature, wetness and maintenance state. BrakeTwin converts approved stopping evidence into conservative deceleration bins for each operating condition.

Step	Method	Safety guard
1. Establish baseline	Controlled stops with mine/OEM approval; record speed, load, grade, surface, brake state and deceleration	No student testing on production vehicle without site approval
2. Create bins	Vehicle class x load x grade x wet/dry x speed band	Sparse bins inherit a worse validated parent value
3. Select lower bound	Use a conservative low quantile/minimum after outlier review	Never use average braking to justify speed
4. Monitor drift	Compare ordinary approved decelerations with expected band	Worsening evidence lowers a_eff immediately
5. Change control	Engineer reviews data and signs new configuration	Evidence cannot automatically raise the speed envelope

Why this is stronger than 'AI safe speed'

- It uses physics for the final decision and data only to set conservative parameters.
- Every recommendation can be reproduced from the log.
- Maintenance drift becomes visible before it turns into an unexplained incident.
- The same algorithm works in the SIH model and later fleet pilot, with different validated bins.

Example only - not a site claim

If a loaded wet-downhill bin has lower effective deceleration than an unloaded dry bin, the advised speed falls even when radar sees the same obstacle at the same range. That behavior is exactly what a fixed red/yellow/green distance zone cannot provide.

DIFFERENTIATOR 2

10. Virtual Block: prevent the blind-curve conflict before it is visible

NEXORA VIRTUAL BLOCK FOR A BLIND SINGLE-LANE SEGMENT



Fail-closed rule Uncertain token state blocks new entry; an occupied vehicle exits conservatively under local radar.

This is inspired by one-train-per-block signalling, adapted as an advisory mine-road control that still requires site validation. [R11]

Token state machine

State	Entry rule	Exit rule / failure
CLEAR	First valid request may be granted after local gate radar/check	Transition to OCCUPIED on confirmed entry
RESERVED	Only granted vehicle/direction may enter before timeout	Cancel only before entry; otherwise manual verification
OCCUPIED	All opposite/new requests held	Clear only on positive exit evidence plus sequence match
UNCERTAIN	No new entry	Existing vehicle exits slowly under local radar; manual clear if evidence is missing
CLOSED	No entry	Supervisor/site procedure required

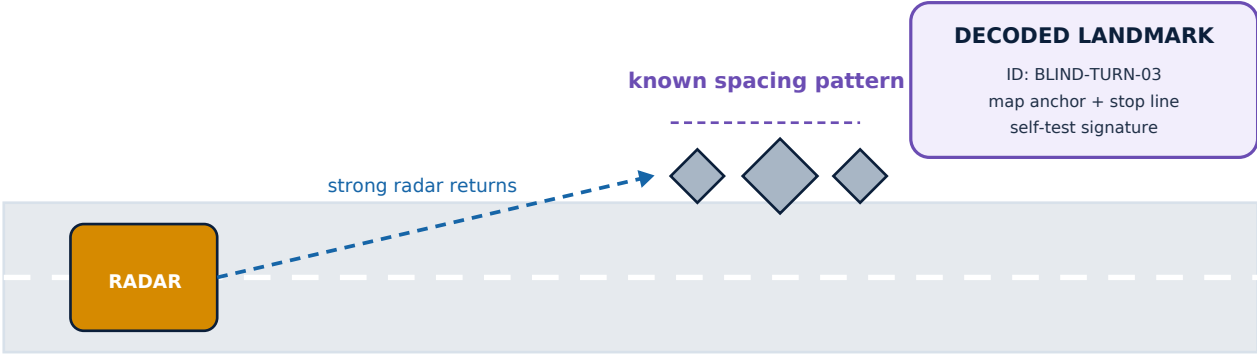
Prototype implementation

- Two ESP32/Wi-Fi or UWB gate nodes, two signal lamps, one edge broker and two scale carts.
- Demonstrate simultaneous opposite requests, dropped messages, delayed exit and power loss.
- Keep it advisory in the HMI; mine traffic rules and physical signs remain authoritative.

DIFFERENTIATOR 3 - RESEARCH HYPOTHESIS

11. RadarGlyph: power-free landmarks for critical fog zones

RADARGLYPH: PASSIVE FOG LANDMARK - RESEARCH MODULE



Hypothesis Passive reflectors may mark GNSS-weak zones; validate pattern detection against clutter.

Radar landmarks have research support; this patterned marker remains an unvalidated prototype. [R10]

Three possible roles

Role	Benefit	Test before claiming
Critical-zone ID	Recognize an approaching blind curve, stop line or bench-edge zone without cloud	Detection/false-match rate across angles, speeds, rain, dust and metal clutter
Map anchor	Correct route-relative pose when GNSS covariance grows	Localization residual against surveyed ground truth
Self-test target	Confirm radar range/angle response during pre-shift/depot pass	Repeatability, temperature drift and contamination sensitivity

Low-cost experiment

- Fabricate 3-4 small metal trihedral reflectors; test individual radar cross-section at several angles.
- Place three reflectors with a distinctive spacing pattern; record range-Doppler-angle signatures.
- Train no black-box model initially: use geometric peak matching with tolerance and negative examples.
- Reject the module if natural mine clutter produces an unacceptable false-ID rate.

Honest positioning

The research basis is radar landmark localization, not proof that Nexora's exact patterned reflector will work. Present RadarGlyph as a low-cost experiment whose value depends on measured detection and false-match results. [R10]

DIFFERENTIATOR 4

12. Proof-carrying cooperative awareness

A peer message should not say only 'truck nearby'. It should carry enough evidence for the receiver to decide whether the information is fresh, route-relevant and physically possible. This makes cooperative awareness auditable and prevents stale GPS dots from becoming silent safety assumptions.

Minimum hazard message

Field	Purpose	Receiver rule
Device identity and authentication	Know the enrolled sender	Reject unknown/invalid signature or MAC
Sequence and timestamp	Detect replay and age	Reject duplicate, backwards or expired packet
Route segment and direction	Map relevance	Ignore safe-speed effect if not on a conflicting corridor
Pose, velocity and covariance	Quantify where the peer could be	Expand predicted occupancy by uncertainty
Predicted path and horizon	Blind-curve conflict	Use bounded 3-5 s route projection, never unlimited extrapolation
Hazard type and evidence source	Explain alert	A peer may add caution; it cannot clear local radar
TTL and validity region	Limit stale influence	Expire to unknown, not safe
Configuration/version	Reproduce behavior	Flag incompatible safety semantics

Communication split

Channel	Use	Never use as
Local wired/CAN	Vehicle state and HMI under approved access	Unapproved brake injection
Direct/site Wi-Fi, private LTE/5G or evaluated V2X	Fresh peer path and virtual-block transactions	Sole local obstacle sensor
LoRa/LoRaWAN	Sparse MOR, node health and low-rate telemetry subject to Indian compliance	Braking-critical, high-rate V2V
Cloud	Long-term analytics and fleet reporting	Real-time local safety loop

MEASURED VISIBILITY FIRST

13. Fog microclimate intelligence without false promises

Fog is often spatially uneven along a mine road. One weather station cannot represent every bench, valley and exposed turn. Nexora should install small nodes only at known fog pockets and decision points, then convert measured visibility into a route-level risk layer.

MOR node inputs

Signal	Role	Claim limit
Forward-scatter/extinction visibility	Primary current visibility estimate	Requires calibration and cleaning; local sample may not represent the full road
Temperature and relative humidity	Dew-point spread and trend	Humidity alone does not prove fog
Rain and surface input	Wet-road BrakeTwin bin and radar clutter context	Does not directly measure friction
Wind	Fog movement/dissipation context	Site-specific relation
Node health	Lens contamination, battery, clock and communications	An unhealthy node cannot raise permitted speed

Nowcast maturity levels

Level	Output	When allowed
L0	Current MOR + health	Day one after calibration
L1	Falling/stable/rising trend	After trend logic is tested
L2	Probability of crossing a visibility threshold in 5-10 min	Only after site-labeled data and calibration curves
L3	Route microclimate forecast	After seasonal validation across nodes

Required wording

Say: 'FogNet measures current visibility and can develop a calibrated local nowcast.' Do not say: 'FogNet always predicts fog 10-20 minutes early.' [R7]

ONE SYSTEM, EXPLICIT DEGRADATION

14. Behavior across environments and failures

Condition	Trusted more	Trusted less	System response
Dense fog	Radar, route graph, MOR	RGB/NIR	Reduce visibility/site cap; keep local radar corridor
Heavy rain	Radar with monitored health, IMU	Camera and raw MOR stability	Wet BrakeTwin bin; monitor clutter/attenuation; tighter buffer
Dust cloud	Radar if radome healthy	Optical classification	Contamination check; conservative class/edge uncertainty
Night	Radar, GNSS/IMU, optional thermal	RGB without illumination	Same stopping envelope; simplify HMI brightness
Glare	Radar and route	Camera confidence	Camera cannot veto radar
GNSS canyon	IMU/wheel/map constraints, optional RadarGlyph	Absolute pose	Grow covariance and reduce route horizon/speed
Network loss	Local radar and onboard model	Peer/MOR freshness	Virtual block fails closed; queue logs
Radar blocked	None can replace primary core	Radar tracks	Immediate degraded/stop request at safe location
Map uncertain	Local radar	Blind-zone and route conflicts	Shorten corridor at uncertainty boundary
Brownout/reboot	Safety MCU and default output	Linux compute until healthy	No spurious normal state; controlled re-entry

Recovery principle

Fault recovery must be slower than fault entry. A single good frame or packet cannot restore normal speed. Require stable evidence for a configured interval, self-test completion and acknowledgement for critical faults.

HUMAN FACTORS

15. Driver HMI: less dashboard, more decision

The Trinetra image is visually rich, but a working dumper HMI should minimize glance time. The driver needs the action and reason, while detailed sensor pages belong in maintenance/control-room views.

Always visible	Only on alert	Maintenance/control room
Actual speed; advised speed; system state; primary health	Direction arrow; hazard/unknown-zone distance; short reason; acknowledgement	Raw tracks; sensor confidence; packet age; logs; calibration; maps; performance charts

Alert grammar

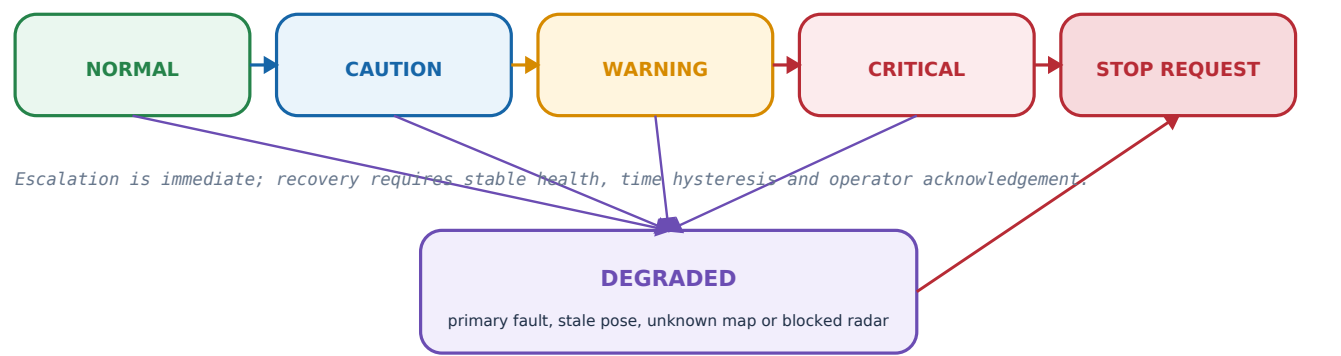
Tier	HMI	Example message	Driver expectation
Attention	Amber visual + single tone	Fog pocket 180 m - target 15 km/h	Prepare and reduce speed
Warning	Amber/red + directional haptic	Stopped vehicle ahead - 42 m	Brake smoothly now
Critical	Red + repeated voice/haptic	Stopping margin negative	Immediate controlled stop
Degraded	Purple + persistent status	Radar blocked - stop at safe place	Follow site recovery procedure

Rules

- Never show an unexplained percentage as the main safety decision.
- One authoritative alert state; suppress duplicate camera/radar/peer alerts for the same conflict.
- Use local languages and cab-noise testing; do not depend only on audio.
- Critical escalation has no debounce delay; recovery uses hysteresis.
- A driver can acknowledge the message, not override a diagnosed primary-sensor fault.

DETERMINISTIC STATE MACHINE

16. Safety states and fail-operational boundary



The SIH build issues advice and a mock output only. Production motion control requires OEM integration and a machine-specific safety case.

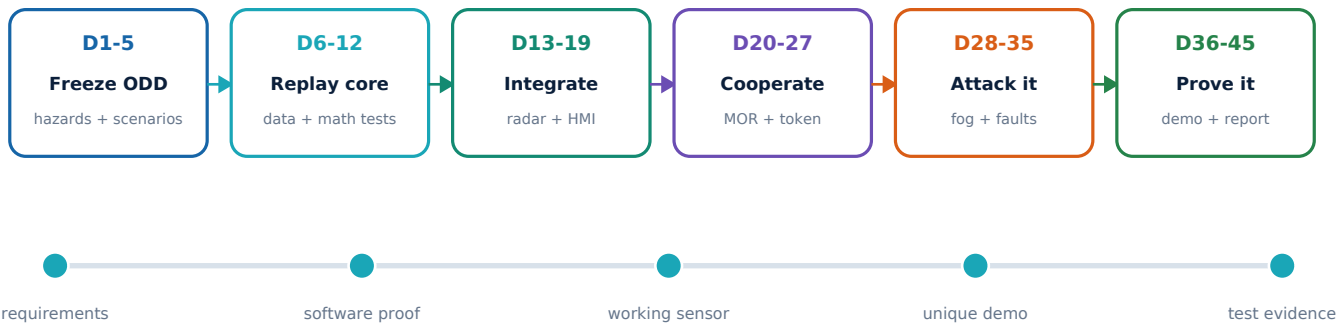
State	Entry example	Speed rule	Exit
Normal	Healthy core; positive reserve	Site/safe-speed minimum	Margin or health worsens
Caution	Low MOR, future blind zone or reduced evidence	Advisory cap with early warning	Stable normal evidence or escalation
Warning	Hazard inside warning envelope	Immediate lower advised speed	Margin restored with hysteresis
Critical	$D_{clear} < D_{req}$ or imminent route conflict	Stop request; loud directional alert	Stopped and conflict cleared
Degraded	Radar/pose/map critical fault	Crawl/stop according to fault and safe-place logic	Self-test + stable health + acknowledgement

Fail-operational vs fail-safe

FogNet may remain **operational at reduced capability** after loss of camera, thermal, MOR node, dashboard or network. It should not claim continued normal operation after loss of the primary radar or trustworthy route localization. In those cases the safe response is degraded travel to a safe place or a stop request, defined with mine operations.

BUILD THE PROOF IN THE CORRECT ORDER

17. 45-day implementation plan



Do not add a new feature after day 35 unless it fixes a measured failure.

Days	Deliverable	Exit criterion
1-5	ODD, route model, hazard log, acceptance metrics and demo storyboard	Team signs one frozen scenario set
6-12	Recorded-data pipeline, golden stopping calculator, unit tests and simulator	Known cases reproduce expected safe speed/state
13-19	Radar integration, tracking, GNSS/IMU, safety MCU heartbeat and simple HMI	Obstacle approach works without network/camera
20-27	MOR node, proof-carrying peer message and Virtual Block prototype	Stale packets rejected; opposite entry held
28-35	Fog/lighting tests, failure injection, RadarGlyph experiment and alert tuning	No missed scripted critical case; defects logged
36-40	Full demo rehearsals, evidence replay, BOM and deployment gate review	Three consecutive clean runs
41-45	Presentation video, final test report, judge Q&A; and spares/recovery plan	A different team member can run the demo

Team workstreams

Lead	Owens
Embedded	Radar ingest, time sync, MCU, power, alerts and hardware health
Algorithms	Tracking, road projection, SafeCorridor, BrakeTwin and tests
Backend/comms	Peer protocol, Virtual Block, store-and-forward and dashboard
Mechanical/test	Mounts, reflectors, MOR fixture, fog rig, ground truth and fault injection
Pitch/safety	ODD, hazard log, metrics, test evidence, cost and judge story

IMPLEMENTATION BLUEPRINT

18. Software design and data contracts

Service	Input	Output	Deadline / guard
Sensor gateway	Radar, GNSS/IMU, vehicle state	Timestamped observations	Monotonic clock; reject stale/out-of-range
Health manager	Diagnostics and signal statistics	Per-source health and usable horizon	Primary fault latched; no silent fallback
Tracker	Radar detections; optional class evidence	Tracks with covariance/provenance	Track expiry and association audit
Route localizer	Pose + graph + optional RadarGlyph	Segment, along-track pose and covariance	Ambiguity shortens horizon
Conflict predictor	Ego/tracks/peers/map	Route occupancy intervals	Bounded prediction; uncertainty expansion
SafeCorridor	Conflicts + health + map horizons	D_clear and limiting reason	Deterministic and unit-tested
BrakeTwin	Class/load/grade/surface/config	a_eff and provenance	No automatic upward update
State manager	Margin, speed caps and faults	One authoritative state	Immediate escalation; hysteretic recovery
HMI/logger	State + reason + evidence	Display/audio/haptic + event capsule	Local operation if network is absent

Event capsule

For every warning/critical/degraded transition, store 10 s before and 20 s after: synchronized radar summary, ego state, route segment, map/config hashes, health, peer packet metadata, MOR, D_clear, D_req, chosen speed, state transition and operator acknowledgement. This enables exact replay and prevents a dashboard screenshot from becoming the only evidence.

Pseudocode

Authoritative loop

```
inputs = acquire_and_timecheck()
health = diagnose(inputs)
tracks = update_tracks(inputs, health)
pose = route_localize(inputs, health)
peers = validate_and_expire(peer_packets)
conflicts = project_to_route(tracks, peers, pose, map)
D_clear, reason = verified_corridor(conflicts, health, pose, map)
a_eff = braking_lower_bound(vehicle, load, grade, surface)
D_req, v_safe = stopping_envelope(D_clear, a_eff, delay, uncertainty)
state = transition(D_clear-D_req, health, hysteresis)
hmi_log_share(state, v_safe, reason, evidence)
```

PROTOTYPE - NOT MINE-CERTIFIED HARDWARE

19. SIH demonstrator BOM and realistic budget

Subsystem	Reference choice	Planning INR	Keep cost low by
Radar	TI IWR6432BOOST 60 GHz evaluation board	18,500 + tax/duty	One board; recorded replay for additional vehicles [R19]
Compute	Raspberry Pi 5, storage and cooling	12,000-20,000	Borrow existing laptop/Pi for first build
Safety MCU	STM32/ESP32 plus watchdog/output	1,500-4,000	Simple independent heartbeat
GNSS + IMU	GNSS module + quality IMU	3,000-10,000	Manual route and surveyed points first
HMI and alerts	5-7 inch display, speaker, haptic, RGB	4,000-8,000	Use laptop/tablet during integration
MOR/weather node	LED/photodiode optical path + T/RH/P/rain	5,000-12,000	One calibrated hotspot node
Virtual Block	2 gate MCUs/radios and signal lamps	3,000-7,000	Use Wi-Fi/ESP-NOW for closed demo
RadarGlyph	Metal corner-reflector prototypes	1,000-4,000	Fabricate locally; test before ruggedizing
Camera	Existing USB/Pi camera	0-4,000	Borrow; evidence only
Power/protection	DC conversion, fuse, transient and E-stop	4,000-10,000	Bench supply for early work
Mechanical/misc	Enclosures, brackets, cables, targets and spares	7,000-15,000	3D print demo mounts

Recommended SIH budget
INR 65,000-1,05,000 for a credible core build if the computer/camera are partly borrowed. Add thermal or RTK only if a tested scenario requires them. Prices are planning ranges, not quotations; verify GST, import duty, availability and lead time before purchase.

Do not publish a INR 7,550 total

A borrowed-part demo may spend that amount in cash, but it is not a reproducible BOM. Judges should see both **cash spent** and **replacement value**.

REAL-WORLD ECONOMICS

20. Closed-track pilot cost, rollout model and ROI

Tier	Scope	Indicative planning range	Not included / caveat
SIH demonstrator	Scale cart or private closed test, one radar, one MOR node, two token gates	INR 0.65-1.05 lakh	Borrowed items and student labor shown separately
Rugged advisory prototype per vehicle	Industrial sensor, edge compute, HMI, harness, enclosure, installation	INR 2.5-5.5 lakh	Not certified; vendor quotes required
Fog hotspot infrastructure	MOR/weather, gate nodes, signs, power and communications	INR 0.5-1.5 lakh per critical site	Civil work and network backhaul vary widely
5-vehicle shadow/advisory pilot	Vehicles, 2-3 hotspots, edge server, engineering and testing	INR 20-45 lakh	Mine access, OEM support, insurance and independent validation extra
Production intervention	OEM retarder/brake interface and safety lifecycle	Quote only after hazard analysis	Do not estimate from hobby BOM

Hotspot-first rollout

- Instrument fog pockets, blind single-lane curves, intersections and bench-edge approaches first.
- Start with two vehicle classes and the worst route; do not blanket-fit the fleet before shadow data.
- Reuse mine private network and surveyed maps where available; local loop must still be independent.

ROI worksheet - use mine data, not invented accident savings

Input	Formula / evidence
Recovered productive hours	Baseline fog shutdown/slowdown hours - pilot hours under the same severity band
Contribution per operating hour	Mine finance figure, not revenue guessed by the team
Avoided disruption	Only documented near-miss/incident/downtime categories agreed with site
Annual system cost	Amortized equipment + network + maintenance + calibration + training + spares
Break-even	Annual system cost / verified contribution per recovered hour

SIX-MINUTE EVIDENCE STORY

21. The SIH live demo that competitors will struggle to copy

Time	Action	What judges see	Proof
0:00-0:35	Normal scale dumper approaches a clear corridor	Green state, D_clear, D_req and advised speed	Deterministic baseline
0:35-1:20	Create fog and hide a stopped target from the camera	Camera confidence drops; radar track remains; safe speed falls	Radar-first local independence
1:20-2:05	Toggle loaded + downhill + wet	Same target range, larger D_req and lower speed	BrakeTwin and physics
2:05-2:50	Two carts request opposite entry to blind curve	One token granted; other held; loss of broker becomes UNCERTAIN	Virtual Block fails closed
2:50-3:30	Disable network/dashboard	Local warning and HMI continue; event queues	Offline-first
3:30-4:10	Replay an old peer packet	Packet rejected by sequence/TTL; reason logged	Proof-carrying message
4:10-4:50	Approach RadarGlyph pattern	Critical-zone ID/map anchor appears; then show negative clutter sample	Research module with measured result
4:50-5:30	Block primary radar	Immediate DEGRADED/stop request; camera cannot pretend normal	Honest failure boundary
5:30-6:00	Open event replay	Exact inputs, config, margin and state transition	Auditability

Physical setup

- Two RC/robot carts, one metal obstacle, one pedestrian cutout/thermal target if used, one curved cardboard/foam wall, two token gates and one MOR node.
- Fog machine/humidifier in a ventilated enclosure; measure visibility. State clearly that lab aerosol is repeatability evidence, not full mine equivalence.
- Large monitor shows only actual speed, advised speed, D_clear, D_req, state and reason; a second engineering view shows raw plots.

The closing line

'We did not add another dashboard. We built a system that can explain exactly why the next metre is safe - and becomes conservative when it cannot.'

TURN CLAIMS INTO EVIDENCE

22. Verification matrix and acceptance targets

ID	Scenario	Fault/variation	Required evidence
T01	Stopped metal vehicle surrogate	Clear/fog/rain spray; multiple ranges	Range error, detection/track continuity, D_clear
T02	Low-reflectivity person surrogate	Lateral offset and night	Detection limits; optional thermal value
T03	Approaching peer at blind curve	Packet delay/loss and GNSS error	Conflict timing and expiry behavior
T04	Virtual Block opposite requests	Broker loss before/after entry	No dual grant; UNCERTAIN behavior
T05	Wet/downhill/load modes	Each BrakeTwin bin	Golden D_req/v_safe agreement
T06	Camera covered	Full loss	Radar warning unchanged; camera health fault
T07	Radar covered/disconnected	Full loss	Degraded/stop request within target
T08	GNSS frozen/jumped	Stale and wrong branch	Innovation fault; covariance and speed cap
T09	MOR node stale	Lost link/bad optical window	No stale visibility clearance; local loop continues
T10	Replay/spoof peer packet	Old sequence/invalid auth	Reject and log; no state reduction
T11	CPU overload/brownout	Missed deadlines/reboot	Watchdog, safe output and controlled recovery
T12	RadarGlyph	Angles, clutter, missing reflector	True/false ID matrix; localization residual

Targets - not claims until passed

Metric	SIH/closed-test target	Report
Critical scripted cases	0 missed hazards across the frozen matrix	Count and replay each case
Local hazard-to-HMI latency	P95 <= 150 ms	Median/P95/max with timestamp method
Golden calculator agreement	Safe speed within 5% or documented numeric tolerance	Unit-test report
Stale peer acceptance	0 packets accepted beyond TTL	Fault-injection log
Virtual Block dual grant	0 across all loss/delay sequences	State/event trace
Nuisance critical alert	<= 1 per test hour after freeze	Cause-coded tally
RadarGlyph detection	Set only after initial feasibility run	Precision/recall by angle and clutter

SAFETY ENGINEERING

23. FMEA: design for the bad day

Failure	Effect	Detection	Safe response	Owner
Radar blocked/misaligned	Primary corridor false/short	Signal statistics, self-test target, residual	Degraded/stop request; service	Vehicle
False radar track	Nuisance braking advice	Track consistency, route gate, replay	Warning with hysteresis; tune, never suppress blindly	Algorithm
Missed low-RCS object	Unseen hazard	Declared test coverage; optional evidence sensor	Lower validated horizon; site barriers/lighting	System/site
GNSS wrong branch	Wrong hazard/grade context	Map innovation and covariance	Shorten corridor; speed cap	Localization
Bad BrakeTwin bin	Unsafe speed recommendation	Config signature, bounds, test review	Worst-class fallback; lock change	Safety lead
Stale peer packet	Ghost/missed conflict	Sequence, time, TTL	Expire to unknown; local radar remains	Comms
Virtual Block lost exit	Road stays blocked	Lease/occupancy discrepancy	Fail closed; manual clear	Operations
Virtual Block false clear	Head-on exposure	Independent entry/exit evidence and audit	No clear without positive exit; critical investigation	System/site
MOR lens dirty	False visibility estimate	Contamination/reference channel	Do not raise cap; maintenance alert	Roadside
HMI/audio failed	Driver misses advice	MCU heartbeat and output test	Degraded indication/stop at safe place	Vehicle
Power/brownout	Uncontrolled reboot	Voltage supervisor/watchdog	Default safe state; no spurious normal	Hardware
Map/config tamper	Wrong speed/geometry	Signature/version and secure boot	Reject and retain last approved config	Cyber

Safety case structure

- Claim: within the declared ODD, FogNet provides timely advisory speed/hazard guidance and degrades visibly.
- Argument: independent local loop, conservative envelope, monitored inputs, controlled configuration and explicit failure states.
- Evidence: requirements trace, unit tests, scenario matrix, latency logs, calibration, FMEA actions, human-factors trial and change records.

A FALSE MESSAGE IS ALSO A SAFETY FAULT

24. Cybersecurity and data integrity

Threat	Control	Safety behavior
Replayed vehicle position	Signed sequence, timestamp, TTL and receiver monotonic cache	Reject; log; do not reduce local alert
Spoofed identity	Provisioned device keys/certificates and allowlist	Treat as unknown object/message
Tampered route/map	Signed artifact, version, hash and two-person release	Reject update; retain last approved map or degrade
Remote alert suppression	Local HMI state owned onboard; remote cannot clear critical	Critical remains until local recovery condition
Denial of service	Rate limits, channel separation and local operation	Peer awareness expires; corridor becomes more conservative
Malicious firmware	Secure boot, signed image, rollback protection	Boot known-good or unavailable
Log editing	Append-only hash chain/event signature and access control	Flag evidence integrity failure
Maintenance misuse	Role-based access, physical service mode and audit	No live calibration change while moving

Network zones

- Safety advisory network: vehicle and gate data, tightly bounded messages.
- Operations network: dashboard, replay, maintenance and approved configuration distribution.
- Enterprise/cloud: long-term analytics only; no dependency for local alerts.

Data minimization

Use device identities rather than driver personal data in the safety message. Keep driver/operator identity only where mine policy and incident investigation require it, with role-based access and retention limits. Follow an OT security lifecycle such as ISA/IEC 62443 for later deployment. [R18]

PROTOTYPE -> PILOT -> PRODUCT

25. Deployment path and standards boundary

Stage	Function	Environment	Gate to advance
S0 simulation	Golden math, replay, faults	Laptop/recorded data	Requirements trace and unit tests
S1 SIH model	Advisory + mock output + token + MOR	Scale carts/private enclosure	Repeatable demo and zero scripted critical misses
S2 closed track	Rugged shadow logging	Non-production vehicle/track	Range, latency, nuisance and environmental evidence
S3 mine shadow	Observe only; compare with driver/site outcomes	2 vehicles, one route, trained observers	ODD coverage and false-negative review
S4 advisory pilot	Driver HMI and hotspot infrastructure	5-10 vehicles	Human factors, availability, maintenance and KPI acceptance
S5 OEM intervention	Retarder/speed inhibit if justified	Named machine type	MCSSA/MPL, ISO 21815/19014, OEM and independent verification
S6 fleet	Managed configuration and support	Operational rollout	Site safety approval, SLA, training and change control

Standards direction

- ISO 21815-1 covers principles, object detection, operator warnings, intervention and tests for earth-moving machinery. It states these systems assist the operator. [R3]
- ISO 19014 addresses functional-safety analysis and performance requirements when machine control is safety-related. [R5, R6]
- The SIH demonstrator should not claim compliance or certification. It should show how its evidence can support a later safety lifecycle.

Production rule

No Raspberry Pi, student relay, remote dashboard command or LoRa packet should be connected to a production dumper's service brake. Any intervention must pass through an OEM-approved architecture and machine-specific validation.

SIH DIFFERENTIATION AND IP HONESTY

26. What is genuinely distinctive - and what is not

Common / prior art	Nexora's distinctive integration
Radar, cameras, thermal, GNSS, maps and proximity alerts	A verified clear corridor compared with a conservative vehicle-specific stopping requirement
Fleet V2V and control-room dashboards	Proof-carrying, route-relevant peer messages that can add risk but cannot clear local evidence
Fog/weather stations	MOR micro-nodes connected directly to route speed caps with health and confidence
Geofencing and blind-turn alerts	Fail-closed occupancy token for defined single-lane fog/blind blocks
Radar landmark localization research	Low-cost patterned passive RadarGlyph hypothesis for mine critical zones and self-test
Vehicle braking models	BrakeTwin bins that may reduce speed from new evidence but cannot auto-increase it
Event logging	Reproducible decision capsule including config/map hashes, state and limiting reason

Potential claim set for a patent professional to examine

- Method for deriving a mine-vehicle Safe-to-Proceed Corridor from route-projected sensor, map, visibility and communication evidence with explicit uncertainty horizons.
- Fail-closed blind-segment reservation integrated with an independent onboard stopping envelope.
- Patterned passive millimeter-wave roadside landmark used jointly for critical-zone identification, route anchoring and radar health self-test.
- Monotonic BrakeTwin governance: adverse evidence may lower the permitted envelope automatically; upward changes require approved validation.

Do not say

'No one has ever thought of this' or 'world's first' without a documented prior-art search. Say: **'We found no identical integrated architecture in our scoped review, and we have isolated four testable differentiators.'**

27. Judge pitch and difficult questions

Eight-slide story

Slide	Message	Visual
1	A vehicle can need more stopping distance than the road it can trust as clear	D_clear vs D_req
2	Sensor-heavy dashboards do not answer safe speed	Competitor gap table
3	Nexora proves a Safe-to-Proceed Corridor	One four-stage architecture
4	BrakeTwin changes speed with load, grade and wetness	Two stopping envelopes
5	Virtual Block handles terrain-hidden conflicts	Blind curve token
6	RadarGlyph and MOR extend trust without cloud	Passive marker + hotspot node
7	Failure demo and evidence replay	Camera/network off; radar continues
8	Advisory pilot now, OEM path later	45-day plan + deployment gates

Expected questions

Question	Answer
How are you different from Trinetra?	Trinetra visualizes sensor detections. Nexora's core output is a proven clear corridor and a vehicle-specific stopping reserve, plus blind-segment fail-closed control.
Is RadarGlyph proven?	No. Radar landmark research supports the direction; our patterned low-cost marker is a measured hypothesis with a rejection criterion.
Why no automatic braking?	The SIH build is advisory. Motion control requires an OEM interface, functional-safety analysis and machine-specific tests under ISO 21815/19014.
Where is AI?	AI may classify objects and nowcast fog. The final safety state is deterministic and replayable.
What if internet/V2V fails?	Local radar and stopping envelope continue. Virtual Block denies new entry if its state is uncertain.
What if radar fails?	We do not pretend another hobby sensor replaces it. The system enters degraded/stop request and follows site recovery rules.
Can fog really be predicted?	We measure current MOR first. Forecast probability is enabled only after locally labeled and calibrated data.
What will you build in SIH?	One working local loop, one MOR node, two token gates, two carts, a replay tool and a complete failure demo.

NEXT 72 HOURS

28. Immediate team action list

Priority	Action	Done when
P0	Remove automatic braking, guaranteed fog prediction, LoRa safety-V2V and Alpha escort claims from the PPT	One consistent architecture in every slide
P0	Freeze one demo ODD and 12-case test matrix	No feature can be added without a scenario/metric
P0	Implement stopping calculator and golden unit tests before hardware	Inputs produce reproducible D_req and v_safe
P1	Select one radar and capture sample frames	Range/velocity tracks replay offline
P1	Build minimal HMI with actual speed, advised speed, margin, state and reason	Readable in under one second
P1	Prototype Virtual Block with packet loss	Never grants both directions
P2	Build/calibrate one MOR node	Current visibility and health logged
P2	Run RadarGlyph feasibility test	Precision/false-match results decide keep/drop
P2	Create event capsule/replay	Every transition is reproducible
P3	Request mentor/mine/OEM review of ODD and braking assumptions	Assumption register signed or corrected

The feature freeze

The build is complete when it proves the core safety claim and survives failures - not when every sensor icon is active.

The sentence every team member should know

Nexora does not ask only whether an object was detected; it asks whether the vehicle can prove a clear stopping corridor, and it becomes conservative whenever that proof weakens.

RESEARCH AND AUTHORITATIVE SOURCES - ACCESSED 19 SEPTEMBER 2026

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TEAM DECISION

Final recommendation

Proceed with **FogNet SafeEnvelope - SafeCorridor Edition** as a radar-first, advisory-first and offline-first system. Make the verified clear corridor and vehicle-specific stopping reserve the product, not the sensor dashboard.

Build this core

Onboard: radar + GNSS/IMU/wheel speed + payload/grade mode + BrakeTwin + deterministic SafeCorridor + safety MCU + minimal HMI.

Roadside: MOR node at one fog hotspot; Virtual Block at one blind segment.

Research stretch: RadarGlyph passive landmarks with a measured keep/drop decision.

Cooperative: authenticated, age-aware route messages; LoRa only for sparse telemetry.

Evidence: fault injection, event capsule replay, acceptance metrics and an honest advisory-to-OEM deployment path.

The three choices that make Nexora credible

1	A network or camera failure never removes a valid local radar warning.
2	New evidence may make the envelope more conservative immediately; it cannot silently justify a higher speed.
3	Every claim is labeled as validated, a target, or a research hypothesis.

Immediate first task: freeze the 12-case test matrix and implement the golden SafeCorridor calculator before purchasing optional sensors.